Fakulteta za informacijske študije
Faculty of information studies

lab04

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Terminal - filip@vmi3135542:~

File Edit View Terminal Tabs Help

```
~
└─ sudo apt install -y gnupg
[sudo] password for filip:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gnupg is already the newest version (2.4.4-2ubuntu17.4).
0 upgraded, 0 newly installed, 0 to remove and 61 not upgraded.
~                                                          ✔ | 7s
└─ gpg --version
gpg (GnuPG) 2.4.4
libgcrypt 1.10.3
Copyright (C) 2024 g10 Code GmbH
License GNU GPL-3.0-or-later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Home: /home/filip/.gnupg
Supported algorithms:
Pubkey: RSA, ELG, DSA, ECDH, ECDSA, EDDSA
Cipher: IDEA, 3DES, CAST5, BLOWFISH, AES, AES192, AES256, TWOFISH,
        CAMELLIA128, CAMELLIA192, CAMELLIA256
Hash: SHA1, RIPEMD160, SHA256, SHA384, SHA512, SHA224
Compression: Uncompressed, ZIP, ZLIB, BZIP2
```

```
~
└─ gpg --full-generate-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: keybox '/home/filip/.gnupg/pubring.kbx' created
Please select what kind of key you want:
   (1) RSA and RSA
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
   (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 1y
Key expires at Thu Apr 22 16:52:50 2027 CEST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: Filip Polic
Email address: filip@gmail.com
Comment: SrbijaSlovenija
You selected this USER-ID:
    "Filip Polic (SrbijaSlovenija) <filip@gmail.com>"
```

```
File  Edit  View  Terminal  Tabs  Help

~                                                                    ✔ | 1m 35s
gpg --list-keys
pg: checking the trustdb
pg: marginals needed: 3  completes needed: 1  trust model: pgp
pg: depth: 0  valid:   1  signed:   0  trust: 0-, 0q, 0n, 0m, 0f, 1u
pg: next trustdb check due at 2027-04-22
home/filip/.gnupg/pubring.kbx
-----------------------------
ub   rsa4096 2026-04-22 [SC] [expires: 2027-04-22]
     148A827197293BCDD0A987F63D5174042FDAE801
id           [ultimate] Filip Polic (SrbijaSlovenija) <filip@gmail.com>
ub   rsa4096 2026-04-22 [E] [expires: 2027-04-22]

~                                                                    ✔
gpg --armor --export student@example.com > student_pubkey.asc
pg: WARNING: nothing exported
~                                                                    ✔
gpg --armor --export filip@gmail.com > student_pubkey.asc
~                                                                    ✔
cat student_pubkey.asc
----BEGIN PGP PUBLIC KEY BLOCK-----

QINBGno4PEBEADFPKOG2v05wN3+uWt48SnBdXzdHd6CozMqBCX+ygoV1gha7Bux
wXAhMnC4Dz7DI0JVC8PnjHkITS8+f2OW+9y6GYlR0gxknUMRpYhPjP7dSAHzpgo
0/UbmPe2QTe2kFGfv2itCuMIwFWSU+u5xTze6I3hwFyjiSfOXjjRc9rUDFpZ000
K4u7qebFGyjoLiQQxOLkbG4+BopJeZdyQiU9w3cgyo11UskytEOhhUJYg2E7B+8
Ud+qCxLCrHq8k+q+Aejb40txGnuKZM64T6PBQ3V9sdhSlXq29bocTkxidbenN9k
JQXNZrWXoD1K6UpIxjlXbQc+iM+o/ZLRfGOi40GTIcoqblK/oDxWZV9BBjQ/MlS
vAcx1NuGYtPxVSjq5PMm0Kxk3ifgwVmt+OLb7SebQuv6Ru7sU38IsNG/AM+OBaL
LkM5HvDvLbDru4SkngTV9cL/YczYSnuqJRQa40+K4uJVUvK+Se/8fMmjCgSmwz/
vNa/TGEfVxKrqMTTg8FdG6kEdI6IndPQsOILYK/+q2dGU9g3QMSYi75So1ZrTAa
NP5YzwtMo76rUkBmJyuFVhMDgB+bDv+pt5N5ilMy09mk0vWPPrfxhI0z5iEkw48
u+xdwGhZpKuSzKLs/ltK0l1fjDTL84546DZbuLqFQSNAOoBLK8FOnF1AQARAQAB
C9GaWxpcCBQb2xpYyAoU3JiaWphU2xvdmVuaWphKSA8ZmlsaXBAZ21haWwuY29t
okCVwQTAQoAQRYhBBSKgnGXKTvN0KmH9j1RdAQv2ugBBQJp6ODxAhsDBQkB4TOA
```

```
~ ✔
└─ echo "To: peer@example.com
From: filip@gmail.com
Date: $(date)
Secret message: Confidential message" > message.txt
~ ✔
└─ cat message.txt
To: peer@example.com
From: filip@gmail.com
Date: Wed Apr 22 17:18:10 CEST 2026
Secret message: Confidential message
~ ✔
└─ gpg --encrypt --sign --armor --recipient peer@example.com --local-user filip@gmail.com
 message.txt
~ ✔ | 13s
└─ cat message.txt.asc
-----BEGIN PGP MESSAGE-----

hQIMA77JcuR+PBBbAQ//b2x1KCN/yxsMfmQLBsP7bsLT1hwRER8n/Npa5PDCyzgh
/+hyjrwHJEXQ0gNf0jrN1eEC2sEnJNrwfyj6HTrVs6bCuCnfHZtfj3O+S3kMVSKo
p0KSgS+cP1KSG6I+eMp5db+I6pgjRW12pLUPj5RV0vo3Lc7DYAz6ykP9iAnd+j2u
tk8ZuvFTFcleIGr7HJVNyX9OxUthC1lAumgc7nbGfdpCkNmfz5DT3yGGFhzVAlSs
lo5KoPAV+fm0L2C10eMLg36AyxKgh/S5jOWHHfd3m2ZRePduehdEy/1uR/4N9Mo7
hvoy2gOE7JxDjsuBPqIkg4VGbSHXZQOiLw+jWTcREvoSVSZdWYNQIe9+VCPEcEe8
fv9MlEy+ey1EucDJA5dvQyFhO7qG1dVNsNMxNGw/VLbd6c+cwbDyz4GuRR020csZ
+AwqZ9+4KQivgHHIRTd99/WWH3wjsBupmgUjGiowc1xOlePhrPKSBrbwIl3VrYVB
1JXNcf+8+vzUecxMNajGPSnpkPnpY0Q8YLGBd+FctHIsHWljqySdkU+/e6V3LCjD
dXOLThFNr3pl7MnicFYQuf90WvuRpgkeaPR9ybiz1ufay9+id10pXBmsOw+dpzGk
0DVhu0VjD3FEfPNYmP8Ak4geuKwTWyO7Rk52NHVAeQZmRtb+nMYobdmFA5C5yurU
6QEJAhBsyQ+Kv51JBPoWgC1JcC3QD5y4YMZ8ptOJlE0fQ9C1lyOik8ptwOu2gGCc
0m3zn5HGAog+ZTw0Tq/heK6HSCJyMjymjejQxfoSzXBuX21D5tJasJ/ZZClePvo9
nzH63XnSMCSwSYUxAlNODjVrv0ykVKe/ljzSwK0qvblEovkbzxfAipat5KvIUs5X
```

```
~ ✔
└─ gpg --decrypt message.txt.asc > decrypted_message.txt
gpg: encrypted with rsa4096 key, ID BEC972E47E3C105B, created 2026-04-22
      "peer@example.com"
gpg: Signature made Wed Apr 22 17:18:30 2026 CEST
gpg:                using RSA key 148A827197293BCDD0A987F63D5174042FDAE801
gpg:                issuer "filip@gmail.com"
gpg: Good signature from "Filip Polic (SrbijaSlovenija) <filip@gmail.com>" [ultimate]
~ ✔
```

```
~
gpg --gen-revoke filip@gmail.com > revoke.asc

sec  rsa4096/3D5174042FDAE801 2026-04-22 Filip Polic (SrbijaSlovenija) <filip@gmail.com>

Create a revocation certificate for this key? (y/N) y
Please select the reason for the revocation:
  0 = No reason specified
  1 = Key has been compromised
  2 = Key is superseded
  3 = Key is no longer used
  Q = Cancel
(Probably you want to select 1 here)
Your decision? 1
Enter an optional description; end it with an empty line:
> NovoMesto00000000000000000
>
Reason for revocation: Key has been compromised
NovoMesto00000000000000000
Is this okay? (y/N) y
ASCII armored output forced.
Revocation certificate created.

Please move it to a medium which you can hide away; if Mallory gets
access to this certificate he can use it to make your key unusable.
It is smart to print this certificate and store it away, just in case
your media become unreadable.  But have some caution:  The print system of
your machine might store the data and make it available to others!
~
✔ | 40s
```

Difference between encryption and signing

Encryption is used to protect the content of a message so that only the intended recipient can read it (confidentiality).
Signing is used to prove who sent the message and to ensure that the content was not changed (authentication and integrity)

Role of public and private key

The public key is shared with others and is used to encrypt messages or verify signatures.
The private key is kept secret and is used to decrypt messages or create a digital signature.

What happens when an encrypted file is modified

If an encrypted file is modified, decryption will usually fail or produce corrupted data.
If the file is also signed, the signature verification will fail, showing that the file was changed.